

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5243A

5 Credits: 4

Program Elective

Source-grounded

Analytical Instruments

□ To impart knowledge on the fundamentals of Analytical instruments

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5243A is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5243A.pdf from the uploaded official SITTTTR syllabus archive.

Course code and title: preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Biomedical Engineering
Course code	5243A
Course title	Analytical Instruments
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4, T:0, P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

11 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To impart knowledge on the fundamentals of Analytical instruments
- To equip students to handle various analytical instruments which are useful for clinical analysis in hospitals and pharmaceutical laboratories.

Course outcomes

CO	Official outcome	Study level
CO1	Describe the fundamentals of Analytical Instrument. 14 Understanding Explain UV, Visible, IR Spectrophotometers and	See official syllabus
CO2	16 Understanding flame photometer Illustrate the instruments for Chromatography,	See official syllabus
CO3	14 Understanding Electrophoresis and densitometry. Explain pH meter, blood cell counters and blood	See official syllabus
CO4	14 Understanding Gas Analyzer	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Describe the fundamentals of Analytical Instruments.	3	As specified
Module 2	Explain UV, Visible, IR Spectrophotometers and flame photometer	2	As specified
Module 3	densitometry.	3	As specified
Module 4	Explain pH meter, blood cell counters and blood Gas Analyzer	3	As specified

MODULE 1

Describe the fundamentals of Analytical Instruments.

This module is presented from the official syllabus scope for Analytical Instruments. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Describe the fundamentals of Analytical Instruments.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ 7 and its performance requirement Summarize the Instrument Calibration

7 and its performance requirement Summarize the Instrument Calibration is an official topic in Module 1 of Analytical Instruments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 7 and its performance requirement Summarize the Instrument Calibration using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 7 and its performance requirement summarize the instrument calibration should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 7 and its performance requirement Summarize the Instrument Calibration means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-7 and its performance requirement Summarize the Instrument Calibration definition/meaning. definition/meaning. steps, application. Technical terms English- Malayalam.

▼ 3 Understanding Techniques & Validation

3 Understanding Techniques & Validation is an official topic in Module 1 of Analytical Instruments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Techniques & Validation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding techniques & validation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding Techniques & Validation means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Understanding Techniques & Validation
 പരീക്ഷയ്ക്കായി definition/meaning നൽകണം. പരീക്ഷയിൽ പരീക്ഷണ
 പരീക്ഷണം, steps, application നൽകണം പരീക്ഷണ പരീക്ഷണം. Technical terms
 English-ൽ പരീക്ഷ പരീക്ഷണം.

▼ **Explain Beer-Lambert's law and colorimeter 4 Understanding Fundamentals of Analytical Instruments Introduction to Analytical Instrumentation: Fundamentals of analytical instruments: Elements of an analytical instrument - PC based analytical instruments - Classification of instrumental techniques. Performance requirement-definition of accuracy, precision, sensitivity, resolution and error, need for calibration Analytical instruments used in clinical environment (listing) and its applications. Electromagnetic radiation - Electromagnetic spectrum- Laws relating to absorption of radiation. Beer-Lambert's Law. Colorimeters - block diagram, Principle & applications, Calibration procedure.**

Explain Beer-Lambert's law and colorimeter 4 Understanding Fundamentals of Analytical Instruments Introduction to Analytical Instrumentation: Fundamentals of analytical instruments: Elements of an analytical instrument - PC based analytical instruments - Classification of instrumental techniques. Performance requirement-definition of accuracy, precision, sensitivity, resolution and error, need for calibration Analytical instruments used in clinical environment (listing) and its applications. Electromagnetic radiation - Electromagnetic spectrum- Laws relating to absorption of radiation. Beer-Lambert's Law. Colorimeters - block diagram, Principle & applications, Calibration procedure. is an official topic in Module 1 of Analytical Instruments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Beer-Lambert's law and colorimeter 4
Understanding Fundamentals of Analytical Instruments Introduction to Analytical Instrumentation: Fundamentals of analytical instruments: Elements of an analytical instrument - PC based analytical instruments - Classification of instrumental techniques. Performance requirement- definition of accuracy, precision, sensitivity, resolution and error, need for calibration Analytical instruments used in clinical environment (listing) and its applications. Electromagnetic radiation - Electromagnetic spectrum- Laws relating to absorption of radiation. Beer-Lambert's Law. Colorimeters - block diagram, Principle & applications, Calibration procedure. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain beer-lambert's law and colorimeter 4 understanding fundamentals of analytical instruments introduction to analytical instrumentation: fundamentals of analytical instruments: elements of an analytical instrument - pc based analytical instruments - classification of instrumental techniques. performance requirement-definition of accuracy, precision, sensitivity, resolution and error, need for calibration analytical instruments used in clinical environment (listing) and its applications. electromagnetic radiation - electromagnetic spectrum- laws relating to absorption of radiation. beer-lambert's law. colorimeters - block diagram, principle & applications, calibration procedure. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Beer-Lambert's law and colorimeter 4 Understanding Fundamentals of Analytical Instruments Introduction to Analytical

Aspect	What to prepare
	Instrumentation: Fundamentals of analytical instruments: Elements of an analytical instrument - PC based analytical instruments - Classification of instrumental techniques. Performance requirement- definition of accuracy, precision, sensitivity, resolution and error, need for calibration Analytical instruments used in clinical environment (listing) and its applications. Electromagnetic radiation - Electromagnetic spectrum- Laws relating to absorption of radiation. Beer-Lambert's Law. Colorimeters - block diagram, Principle & applications, Calibration procedure. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain Beer-Lambert's law and colorimeter 4 Understanding Fundamentals of Analytical Instruments Introduction to Analytical Instrumentation: Fundamentals of analytical instruments: Elements of an analytical instrument - PC based analytical instruments - Classification of instrumental techniques. Performance requirement-definition of accuracy, precision, sensitivity, resolution and error, need for calibration Analytical instruments used in clinical environment (listing) and its applications. Electromagnetic radiation - Electromagnetic spectrum- Laws relating to absorption of radiation. Beer-Lambert's Law. Colorimeters - block diagram, Principle & applications, Calibration procedure. definition/meaning

പ്രവർത്തനം. പദങ്ങൾ പദങ്ങൾ പദങ്ങൾ, steps, application പദങ്ങൾ
പദങ്ങൾ പദങ്ങൾ. Technical terms English-പദങ്ങൾ പദങ്ങൾ.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Explain UV, Visible, IR Spectrophotometers and flame photometer

This module is presented from the official syllabus scope for Analytical Instruments. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain UV, Visible, IR Spectrophotometers and flame photometer
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

▼ 9 Understanding and IR spectrophotometers Illustrate the working principle and

9 Understanding and IR spectrophotometers Illustrate the working principle and is an official topic in Module 2 of Analytical Instruments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 9 Understanding and IR spectrophotometers Illustrate the working principle and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 9 understanding and ir spectrophotometers illustrate the working principle and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 9 Understanding and IR spectrophotometers Illustrate the working principle and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-9 Understanding and IR spectrophotometers Illustrate the working principle and definition/meaning. steps, application. Technical terms English- Malayalam.

▼ 2 Understanding constructional details of flame photometer. SPECTROPHOTOMETER Principle of absorption spectroscopy- Types(Listing) Spectrophotometers: basic sub components - monochromator, detection systems and amplifiers - single beam & double beam - visible, UV & IR (basics only), - applications in biochemical analysis. Flame Photometry: Principle and constructional details of flame photometer- Emission system - Optical system - Detectors. Applications of flame photometry. Illustrate the instruments for Chromatography, Electrophoresis and

2 Understanding constructional details of flame photometer.

SPECTROPHOTOMETER Principle of absorption spectroscopy- Types(Listing) Spectrophotometers: basic sub components - monochromator, detection systems and amplifiers - single beam & double beam - visible, UV & IR (basics only), - applications in biochemical analysis. Flame Photometry: Principle and constructional details of flame photometer- Emission system - Optical system - Detectors. Applications of flame photometry. Illustrate the instruments for Chromatography, Electrophoresis and is an official topic in Module 2 of Analytical Instruments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding constructional details of flame photometer. SPECTROPHOTOMETER Principle of absorption spectroscopy- Types(Listing) Spectrophotometers: basic sub components - monochromator, detection systems and amplifiers - single beam & double

beam - visible, UV & IR (basics only), - applications in biochemical analysis.

Flame Photometry: Principle and constructional details of flame photometer-

Emission system - Optical system - Detectors. Applications of flame photometry. Illustrate the instruments for Chromatography, Electrophoresis and using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding constructional details of flame photometer. spectrophotometer principle of absorption spectroscopy- types(listing) spectrophotometers: basic sub components - monochromator, detection systems and amplifiers - single beam & double beam - visible, uv & ir (basics only), - applications in biochemical analysis. flame photometry: principle and constructional details of flame photometer- emission system - optical system - detectors. applications of flame photometry. illustrate the instruments for chromatography, electrophoresis and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding constructional details of flame photometer. SPECTROPHOTOMETER Principle of absorption spectroscopy- Types(Listing) Spectrophotometers: basic sub components - monochromator, detection systems and amplifiers - single beam & double beam - visible, UV & IR (basics only), - applications in biochemical analysis. Flame Photometry: Principle and constructional details of flame photometer- Emission system - Optical system - Detectors. Applications of flame photometry. Illustrate the instruments for Chromatography, Electrophoresis and means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 2 Understanding constructional details of flame photometer. SPECTROPHOTOMETER Principle of absorption spectroscopy- Types(Listing) Spectrophotometers: basic sub components - monochromator, detection systems and amplifiers - single beam & double beam - visible, UV & IR (basics only), - applications in biochemical analysis. Flame Photometry: Principle and constructional details of flame photometer- Emission system - Optical system - Detectors. Applications of flame photometry. Illustrate the instruments for Chromatography, Electrophoresis and definition/meaning. steps, application. Technical terms English- Malayalam.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3**densitometry.**

This module is presented from the official syllabus scope for Analytical Instruments. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: densitometry.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ Explain chromatography and its types. 3 Understanding Explain the principle and construction details of

Explain chromatography and its types. 3 Understanding Explain the principle and construction details of is an official topic in Module 3 of Analytical Instruments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain chromatography and its types. 3 Understanding Explain the principle and construction details of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain chromatography and its types. 3 understanding explain the principle and construction details of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain chromatography and its types. 3 Understanding Explain the principle and construction details of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Explain chromatography and its types. 3 Understanding Explain the principle and construction details of definition/meaning. steps, application. Technical terms English- .

▼ 6 gas and liquid chromatography in detail. Understanding Describe the working of electrophoresis machine

6 gas and liquid chromatography in detail. Understanding Describe the working of electrophoresis machine is an official topic in Module 3 of Analytical Instruments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 gas and liquid chromatography in detail. Understanding Describe the working of electrophoresis machine using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 gas and liquid chromatography in detail. understanding describe the working of electrophoresis machine should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 gas and liquid chromatography in detail. Understanding Describe the working of electrophoresis machine means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 6 gas and liquid chromatography in detail. Understanding Describe the working of electrophoresis machine definition/meaning, steps, application Technical terms English-

▼ 5 Understanding and densitometer Chromatography And Electrophoresis Chromatographic process - Classification- Terms in chromatography - Gas chromatography: Block diagram - Principle - Constructional details Liquid Chromatography: Types of liquid chromatography- High pressure Liquid Chromatography (HPLC): Principle - Constructional details, Applications in medical field. Electrophoresis apparatus - basic principle and block diagram. Densitometer - principle and block diagram

5 Understanding and densitometer Chromatography And Electrophoresis Chromatographic process - Classification- Terms in chromatography - Gas chromatography: Block diagram - Principle - Constructional details Liquid Chromatography: Types of liquid chromatography- High pressure Liquid Chromatography (HPLC): Principle - Constructional details, Applications in medical field. Electrophoresis apparatus - basic principle and block diagram. Densitometer - principle and block diagram is an official topic in Module 3 of Analytical Instruments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding and densitometer Chromatography And Electrophoresis Chromatographic process - Classification- Terms in chromatography - Gas chromatography: Block diagram - Principle - Constructional details Liquid Chromatography: Types of liquid chromatography- High pressure Liquid Chromatography (HPLC): Principle - Constructional details, Applications in medical field. Electrophoresis

apparatus - basic principle and block diagram. Densitometer - principle and block diagram using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding and densitometer chromatography and electrophoresis chromatographic process - classification- terms in chromatography - gas chromatography: block diagram - principle - constructional details liquid chromatography: types of liquid chromatography- high pressure liquid chromatography (hplc): principle - constructional details, applications in medical field. electrophoresis apparatus - basic principle and block diagram. densitometer - principle and block diagram should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding and densitometer Chromatography And Electrophoresis Chromatographic process - Classification- Terms in chromatography - Gas chromatography: Block diagram - Principle - Constructional details Liquid Chromatography: Types of liquid chromatography- High pressure Liquid Chromatography (HPLC): Principle - Constructional details, Applications in medical field. Electrophoresis apparatus - basic principle and block diagram. Densitometer - principle and block diagram means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-5 Understanding and densitometer Chromatography And Electrophoresis Chromatographic process - Classification- Terms in chromatography - Gas chromatography: Block diagram - Principle - Constructional details Liquid Chromatography: Types of liquid chromatography- High pressure Liquid Chromatography (HPLC): Principle - Constructional details, Applications in medical field. Electrophoresis apparatus - basic principle and block diagram. Densitometer - principle and block diagram

 definition/meaning
 .
 , steps, application
 . Technical terms English-
 .

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Explain pH meter, blood cell counters and blood Gas Analyzer

This module is presented from the official syllabus scope for Analytical Instruments. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain pH meter, blood cell counters and blood Gas Analyzer
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ 3 Understanding meter. Detail the working of blood cell counters and

3 Understanding meter. Detail the working of blood cell counters and is an official topic in Module 4 of Analytical Instruments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding meter. Detail the working of blood cell counters and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding meter. detail the working of blood cell counters and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 3 Understanding meter. Detail the working of blood cell counters and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 3 Understanding meter. Detail the working of blood cell counters and definition/meaning. steps, application. Technical terms English- .

▼ 6 Understanding Hb meter. Describe the principle & construction of

6 Understanding Hb meter. Describe the principle & construction of is an official topic in Module 4 of Analytical Instruments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding Hb meter. Describe the principle & construction of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding hb meter. describe the principle & construction of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 6 Understanding Hb meter. Describe the principle & construction of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-6 Understanding Hb meter. Describe the principle & construction of ഹീമറ്റർ definition/meaning ഹീമറ്റർ. ഹീമറ്റർ ഹീമറ്റർ ഹീമറ്റർ, steps, application ഹീമറ്റർ ഹീമറ്റർ ഹീമറ്റർ. Technical terms English-ഹീമറ്റർ ഹീമറ്റർ.

▼ **5 Understanding Blood Gas Analyzer and glucometer pH METER, BLOOD CELL COUNTERS & BLOOD GAS ANALYZER pH meter - Principle of pH measurement - Electrodes for pH measurement - Block diagram of pH meter. Types of blood cells, Blood cell counters - types - coulter counter and flow cytometry. Auto analyzers - block diagram and working. Hb meter - block diagram Blood Gas Analyzer - pO₂ and pCO₂ measurement - Principle, construction and working of PO₂ electrode and PCO₂ electrode - Blood Gas Analyzer - block diagram of blood gas analyser. Glucometer - working principle and basic block diagram Text / Reference: T/R Book Title/Author R.S. Khandpur, 'Handbook of Analytical Instruments', Tata McGraw Hill T1 publishing Co. Ltd., 2003. T2 G.W. Ewing, Instrumental Methods of Analysis, Mc Graw Hill, 2004. R1 Robrt D Braun', 'Introduction to Instrumental Analysis', Mc Graw Hill Ltd Galen W Ewing, 'Instrumental methods of chemical analysis', Mc Graw Hill R2 Ltd Online Resources: Sl.No Website Link 1 <https://www.hemakumar.in/analytical-instrumentation> 2 <https://www.slideshare.net/ShreyaAhuja2/ph-meter-54084084> 3 <https://www.ktuassist.in/2020/06/ktu-ae402-analytical-instrumentation.html#>**

5 Understanding Blood Gas Analyzer and glucometer pH METER, BLOOD CELL COUNTERS & BLOOD GAS ANALYZER pH meter - Principle of pH measurement - Electrodes for pH measurement - Block diagram of pH meter. Types of blood cells, Blood cell counters - types - coulter counter and flow cytometry. Auto analyzers - block diagram and working. Hb meter - block diagram Blood Gas Analyzer - pO₂ and pCO₂ measurement - Principle, construction and working of PO₂ electrode and PCO₂ electrode - Blood Gas Analyzer - block diagram of blood gas analyser. Glucometer - working principle and basic block diagram Text / Reference: T/R Book Title/Author R.S. Khandpur, 'Handbook of Analytical Instruments', Tata McGraw Hill T1 publishing Co. Ltd., 2003. T2 G.W. Ewing, Instrumental Methods of Analysis, Mc Graw Hill, 2004. R1 Robrt D Braun', 'Introduction to Instrumental Analysis', Mc Graw Hill Ltd Galen W Ewing, 'Instrumental methods of chemical analysis', Mc Graw Hill R2 Ltd Online Resources:

Sl.No Website Link 1 <https://www.hemakumar.in/analytical-instrumentation>
2 <https://www.slideshare.net/ShreyaAhuja2/ph-meter-54084084> 3
<https://www.ktuassist.in/2020/06/ktu-ae402-analytical-instrumentation.html#> is an official topic in Module 4 of Analytical Instruments. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding Blood Gas Analyzer and glucometer pH METER, BLOOD CELL COUNTERS & BLOOD GAS ANALYZER pH meter - Principle of pH measurement - Electrodes for pH measurement - Block diagram of pH meter. Types of blood cells, Blood cell counters - types - coulter counter and flow cytometry. Auto analyzers - block diagram and working. Hb meter - block diagram Blood Gas Analyzer - pO₂ and pCO₂ measurement - Principle, construction and working of PO₂ electrode and PCO₂ electrode - Blood Gas Analyzer - block diagram of blood gas analyser. Glucometer - working principle and basic block diagram Text / Reference: T/R Book Title/Author R.S. Khandpur, 'Handbook of Analytical Instruments', Tata McGraw Hill T1 publishing Co. Ltd., 2003. T2 G.W. Ewing, Instrumental Methods of Analysis, Mc Graw Hill, 2004. R1 Robrt D Braun', 'Introduction to Instrumental Analysis', Mc Graw Hill Ltd Galen W Ewing, 'Instrumental methods of chemical analysis', Mc Graw Hill R2 Ltd Online Resources: Sl.No Website Link 1 <https://www.hemakumar.in/analytical-instrumentation> 2 <https://www.slideshare.net/ShreyaAhuja2/ph-meter-54084084> 3 <https://www.ktuassist.in/2020/06/ktu-ae402-analytical-instrumentation.html#> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.

- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding blood gas analyzer and glucometer ph meter, blood cell counters & blood gas analyzer ph meter - principle of ph measurement - electrodes for ph measurement - block diagram of ph meter. types of blood cells, blood cell counters - types - coulter counter and flow cytometry. auto analyzers - block diagram and working. hb meter - block diagram blood gas analyzer - po2 and pco2 measurement - principle, construction and working of po2 electrode and pco2 electrode - blood gas analyzer - block diagram of blood gas analyser. glucometer - working principle and basic block diagram text / reference: t/r book title/author r.s. khandpur, 'handbook of analytical instruments', tata mcgraw hill t1 publishing co. ltd., 2003. t2 g.w. ewing, instrumental methods of analysis, mc grow hill, 2004. r1 robt d braun', 'introduction to instrumental analysis', mc grow hill ltd galen w ewing, 'instrumental methods of chemical analysis', mc grow hill r2 ltd online resources: sl.no website link 1 <https://www.hemakumar.in/analytical-instrumentation> 2 <https://www.slideshare.net/shreyaahuja2/ph-meter-54084084> 3 <https://www.ktuassist.in/2020/06/ktu-ae402-analytical-instrumentation.html#> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding Blood Gas Analyzer and glucometer pH METER, BLOOD CELL COUNTERS & BLOOD GAS ANALYZER pH meter - Principle of pH measurement - Electrodes for pH measurement - Block diagram of pH meter. Types of blood cells, Blood cell counters - types - coulter counter and flow cytometry. Auto analyzers - block diagram and working. Hb meter - block diagram Blood Gas Analyzer - pO ₂ and pCO ₂ measurement - Principle, construction and working of PO ₂ electrode and PCO ₂ electrode - Blood Gas Analyzer - block diagram of blood gas analyser. Glucometer - working principle and basic block diagram Text / Reference: T/R Book Title/Author R.S. Khandpur, 'Handbook of Analytical

Aspect	What to prepare
	Instruments', Tata McGraw Hill T1 publishing Co. Ltd., 2003. T2 G.W. Ewing, Instrumental Methods of Analysis, Mc Graw Hill, 2004. R1 Robrt D Braun', 'Introduction to Instrumental Analysis', Mc Graw Hill Ltd Galen W Ewing, 'Instrumental methods of chemical analysis', Mc Graw Hill R2 Ltd Online Resources: Sl.No Website Link 1 https://www.hemakumar.in/analytical-instrumentation 2 https://www.slideshare.net/ShreyaAhuja2/ph-meter-54084084 3 https://www.ktuassist.in/2020/06/ktu-ae402-analytical-instrumentation.html# means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-5 Understanding Blood Gas Analyzer and glucometer pH METER, BLOOD CELL COUNTERS & BLOOD GAS ANALYZER pH meter - Principle of pH measurement - Electrodes for pH measurement - Block diagram of pH meter. Types of blood cells, Blood cell counters - types - coulter counter and flow cytometry. Auto analyzers - block diagram and working. Hb meter - block diagram Blood Gas Analyzer - pO₂ and pCO₂ measurement - Principle, construction and working of PO₂ electrode and PCO₂ electrode - Blood Gas Analyzer - block diagram of blood gas analyser. Glucometer - working principle and basic block diagram Text / Reference: T/R Book Title/Author R.S. Khandpur, 'Handbook of Analytical Instruments', Tata McGraw Hill T1 publishing Co. Ltd., 2003. T2 G.W. Ewing, Instrumental Methods of Analysis, Mc Graw Hill, 2004. R1 Robrt D Braun', 'Introduction to Instrumental Analysis', Mc Graw Hill Ltd Galen W Ewing, 'Instrumental methods of chemical analysis', Mc Graw Hill R2 Ltd Online Resources: Sl.No

Website Link 1 <https://www.hemakumar.in/analytical-instrumentation> 2
<https://www.slideshare.net/ShreyaAhuja2/ph-meter-54084084> 3
<https://www.ktuassist.in/2020/06/ktu-ae402-analytical-instrumentation.html#> ഫീഡ്ബാക്ക് definition/meaning
 ഫീഡ്ബാക്ക്. ഫീഡ്ബാക്ക് ഫീഡ്ബാക്ക്, steps, application ഫീഡ്ബാക്ക്
 ഫീഡ്ബാക്ക് ഫീഡ്ബാക്ക്. Technical terms English-ഫീഡ്ബാക്ക് ഫീഡ്ബാക്ക്.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle →
 Components/steps →
 Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
 → Observation → Result →
 Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL**Module-wise questions and answers**

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► **Define or explain 7 and its performance requirement Summarize the Instrument Calibration. (3 marks)**

► **Define or explain 3 Understanding Techniques & Validation. (3 marks)**

► **Define or explain Explain Beer-Lambert's law and colorimeter 4 Understanding Fundamentals of Analytical Instruments Introduction to Analytical Instrumentation: Fundamentals of analytical instruments: Elements of an analytical instrument - PC based analytical instruments - Classification of instrumental techniques. Performance requirement- definition of accuracy, precision, sensitivity, resolution and error, need for calibration Analytical instruments used in clinical environment (listing) and its applications. Electromagnetic radiation - Electromagnetic spectrum- Laws relating to absorption of radiation. Beer-Lambert's Law. Colorimeters - block diagram, Principle & applications, Calibration procedure.. (3 marks)**

Module 2

► **Define or explain 9 Understanding and IR spectrophotometers Illustrate the working principle and. (3 marks)**

► **Define or explain 2 Understanding constructional details of flame photometer. SPECTROPHOTOMETER Principle of absorption spectroscopy- Types(Listing) Spectrophotometers: basic sub**

components - monochromator, detection systems and amplifiers - single beam & double beam - visible, UV & IR (basics only), - applications in biochemical analysis. Flame Photometry: Principle and constructional details of flame photometer- Emission system - Optical system - Detectors. Applications of flame photometry. Illustrate the instruments for Chromatography, Electrophoresis and. (3 marks)

Module 3

► **Define or explain Explain chromatography and its types. 3 Understanding Explain the principle and construction details of. (3 marks)**

► **Define or explain 6 gas and liquid chromatography in detail. Understanding Describe the working of electrophoresis machine. (3 marks)**

► **Define or explain 5 Understanding and densitometer Chromatography And Electrophoresis Chromatographic process - Classification- Terms in chromatography - Gas chromatography: Block diagram - Principle - Constructional details Liquid Chromatography: Types of liquid chromatography- High pressure Liquid Chromatography (HPLC): Principle - Constructional details, Applications in medical field. Electrophoresis apparatus - basic principle and block diagram. Densitometer - principle and block diagram. (3 marks)**

Module 4

► **Define or explain 3 Understanding meter. Detail the working of blood cell counters and. (3 marks)**

► **Define or explain 6 Understanding Hb meter. Describe the principle & construction of. (3 marks)**

► **Define or explain 5 Understanding Blood Gas Analyzer and glucometer pH METER, BLOOD CELL COUNTERS & BLOOD GAS ANALYZER pH meter - Principle of pH measurement - Electrodes for pH measurement - Block diagram of pH meter. Types of blood cells, Blood cell counters - types - coulter counter and flow cytometry. Auto analyzers - block diagram and working. Hb meter - block diagram Blood Gas Analyzer - pO₂ and pCO₂ measurement - Principle, construction and working of PO₂ electrode and PCO₂ electrode - Blood Gas Analyzer - block diagram of blood gas analyser. Glucometer - working principle and basic block diagram Text / Reference: T/R Book Title/Author R.S. Khandpur, 'Handbook of Analytical Instruments', Tata McGraw Hill T1 publishing Co. Ltd., 2003. T2 G.W. Ewing, Instrumental Methods of Analysis, Mc Graw Hill, 2004. R1 Robrt D Braun' , 'Introduction to Instrumental Analysis' , Mc Graw Hill Ltd Galen W Ewing, 'Instrumental methods of chemical analysis' , Mc Graw Hill R2 Ltd Online Resources: Sl.No Website Link 1 <https://www.hemakumar.in/analytical-instrumentation> 2 <https://www.slideshare.net/ShreyaAhuja2/ph-meter-54084084> 3 <https://www.ktuassist.in/2020/06/ktu-ae402-analytical-instrumentation.html#>. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **7 and its performance requirement Summarize the Instrument Calibration.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **9 Understanding and IR spectrophotometers Illustrate the working principle and.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Explain chromatography and its types. 3 Understanding Explain the principle and construction details of.**

Module 4

1-mark definition: State the meaning of **3 Understanding meter. Detail the working of blood cell counters and.**

3-mark explanation: Explain one principle, process or comparison from

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- <https://www.hemakumar.in/analytical-instrumentation>
- <https://www.slideshare.net/ShreyaAhuja2/ph-meter-54084084>
- <https://www.ktuassist.in/2020/06/ktu-ae402-analytical-instrumentation.html#>

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 5243A. Verify institutional updates against the current official curriculum.